

Concept-Space BM25: When Does BM25-Style Scoring Beat Inner-Product Scoring over Learned Concept Vocabularies?

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Abstract

BM25 is a strong, fully interpretable ranking function, conventionally defined over discrete lexical terms. Whether its mechanisms transfer to a learned semantic vocabulary is no longer open: four independent 2026 works answer yes. What none answers is *when* — each fixes one concept basis and one scoring function, reports that the pair works, and leaves the gain unattributable. We propose a controlled **representation** $\times$ **scoring** factorial. Concept-activation representations are built from a frozen dense encoder using seven bases (identity, PCA, random projection, k -means, product and residual quantisation, sparse autoencoders), at pooled and token granularity, and scored with both inner product and a BM25-style function whose components are individually ablatable. Every configuration is instrumented with concept-distribution diagnostics — Zipf slope, IDF variance, effective vocabulary size, Gini, live-feature fraction — and the relation between those diagnostics and the BM25-minus-inner-product margin is the primary object of study rather than a sanity check. We also prove that a purely linear concept-space score collapses to a bilinear form on the original embeddings, and pre-register an analysis plan in which a null is as informative as a win.

1 Introduction

BM25 weights rare features heavily, saturates repeated evidence, and discounts long documents. None of these mechanisms is inherently lexical: each needs only non-negative features with meaningful collection frequencies, magnitudes and lengths — structure raw dense embeddings lack and learned sparse representations supply. Four groups demonstrated this independently between March and May 2026, over SAE features from a frozen dense retriever (Clavié et al., 2026), over SAE visual words from ViT patches (Han et al.,

Representation	Inner prod.	BM25
SPLADE-v3 (MLM vocab.)	0.568	0.455
SAE feats. over Nomic	0.381	0.582

Table 1: The scoring-function crossover (Clavié et al., 2026): nDCG@10 over selected BEIR datasets.

2026), and as the output vocabulary for SPLADE-style training (Formal et al., 2026; Zong et al., 2026). Asking whether BM25 can score semantic concepts would be answering a settled question.

The hole they leave is elsewhere: each fixes one basis and one scorer, and none varies the basis holding the scorer fixed, or the reverse. Nobody can predict, for a new representation, whether BM25-style scoring will help or hurt — and Table 1 shows the answer flips. Two sparse, non-negative, high-dimensional representations; the best scorer for one is the worst for the other, by 0.113 and 0.201 nDCG@10.

The scoring function is not a matter of convention. Clavié et al. (2026) explain the reversal — SPLADE’s FLOPs regularisation pushes its activation distribution away from the Zipfian shape BM25 assumes, while SAE features retain it — but test that on one pair of models. It is a hypothesis stated once, at $n = 2$.

Central question. Under what statistical properties of a learned concept vocabulary does BM25-style scoring beat inner-product scoring on the *same* representation, and which of BM25’s components carries the difference?¹

Three things follow. The concept basis becomes the primary independent variable; the diagnostics become explanatory variables rather than prelimi-

¹“Semantic BM25” is taken by Lu et al. (2025) for what a cross-encoder computes internally — mechanistic interpretability, not a retrieval method, so no priority conflict, but a reviewer would notice. We use **Concept-Space BM25** (S-BM25) and cite that work as the emergent counterpart to our imposed construction.

naries; and a negative result becomes publishable, since “BM25-style scoring fails when concept frequencies are near-uniform, and here is the threshold” is a finding where “our method did not beat cosine” is not.

2 Background

For query Q and document D ,

$$\text{BM25}(Q, D) = \sum_{t \in Q} \text{IDF}(t) \frac{f(t, D)(k_1 + 1)}{f(t, D) + k_1 \left(1 - b + b \frac{|D|}{\text{avg}|D|}\right)} \quad (1)$$

with $\text{IDF}(t) = \log \frac{N - n(t) + 0.5}{n(t) + 0.5}$. Many mutually incompatible formulas circulate under this name (Kamphuis et al., 2020); we fix the Okapi TREC-3 form (Robertson et al., 1995) and report it explicitly. Four mechanisms separate out: (M1) rarity weighting, (M2) concave bounded saturation in f , (M3) length normalisation via b , (M4) sparse support, which makes an inverted index natural. On a raw dense vector all four break at once: components are signed, so f is not a frequency; every coordinate fires in every document, so $n(t) = N$ and IDF is a negative constant; there is no length and no sparse support. Every construction that works first manufactures a sparse, non-negative, count-like structure — and *how* it is manufactured is what we study.

Among prior sparse retrievers, COILcr (Fan et al., 2023) clusters COIL’s (Gao et al., 2021) term vectors by spherical k -means — a precedent for inducing a retrieval vocabulary by clustering. Two prior findings shape our design: randomly projected sparse vocabularies are markedly worse than learned ones (Zamani et al., 2018; Zong et al., 2026), so random projection enters as a designed control rather than a candidate; and structured SAE variants show no retrieval advantage over plain Top- K (Gao et al., 2025) in two independent reports (Zong et al., 2026; Clavié et al., 2026), licence to fix Top- K and skip architecture search.

3 Related Work and the Gap

Organised by representation and by scorer, the field has conspicuous holes. Inner-product scoring covers token vectors in inverted lists (Gao et al., 2021; Fan et al., 2023), quantised codes (Jégou et al., 2011; Zhu et al., 2024) and projected sparse vocabularies (Zamani et al., 2018); BM25-style scoring has been tried only on MLM vocabularies, where it degrades (Clavié et al., 2026), and on token- and patch-level SAE features, where it helps (Clavié

et al., 2026; Han et al., 2026). Every quantisation, clustering and projection basis is unstudied under BM25 scoring, and pooled SAE features under both. That BM25-on-vectors predates neural IR by two decades — Video Google (Sivic and Zisserman, 2003) indexing quantised SIFT descriptors, that line rediscovering saturation unaided (Jégou et al., 2009; Zheng et al., 2018) — is evidence the mechanisms are representation-general rather than language-specific.

The pooled-versus-token confound. CL-SR (Park et al., 2025) trains an SAE on the *pooled* single vector and scores SPLADE-style, falling below its own backbone (MRR@10 36.8 vs. SimLM’s 41.1, per Zong et al. (2026)); Latent Terms (Clavié et al., 2026) applies the SAE at *token* level and sum-pools after, beating its backbone (0.415 $\rightarrow$ 0.474 BEIR nDCG@10). The nominal difference is granularity — but the two also differ in scoring function, SAE training corpus and backbone, so the natural reading is not identified.

Codebook utilisation, unconnected. Generative retrieval measures what our diagnostics measure without naming it — hourglass concentration of intermediate residual-quantised codes (Zhu et al., 2024), effective codebook capacity and codeword overlap (Wang et al., 2026) — since code-usage imbalance is, from BM25’s side, what decides whether IDF can discriminate. That field treats imbalance as a defect to minimise; BM25 treats it as a precondition. Both measure one statistic with opposite sign conventions.

Gaps. **G1** No factorial: every published result is a single (representation, scorer) point. **G2** The Zipf hypothesis has never been varied as a controlled variable. **G3** Pooled vs. token is confounded with three other factors. **G4** k -means, PQ, RQ and PCA have indexed dense embeddings but never served as BM25 vocabularies. **G5** No component-level ablation, so which mechanism carries the gain is unknown.

4 Method

Concept bases. A *concept basis* is a map $\Phi : \mathbb{R}^d \rightarrow \mathbb{R}_{\geq 0}^K$; we require only non-negativity and a defined K . Non-negativity is load-bearing, making x_{dj} readable as a frequency and keeping length meaningful. Bases emitting signed output are split-rectified, $\Phi^\pm(h) = [\text{ReLU}(Wh); \text{ReLU}(-Wh)]$,

preserving all information — unlike a bare ReLU — at the price of doubling nominal vocabulary size.

Granularity. Under **P** (pooled), $x_d = \phi(\Phi(e_d))$; under **T** (token), $x_d = \phi(\sum_i \Phi(h_i))$. This is the crux of G3: under **P**, L_d says nothing about token count, so we expect — and test — that length normalisation is inert there and active under **T**.

Activation transform. Summed SAE activations are magnitudes, not counts, so following Clavié et al. (2026) we apply $\phi(u) = u^\alpha$, default $\alpha = \frac{1}{2}$. This has a second function: it reshapes the frequency distribution, making α a within-basis lever on the statistic our diagnostics measure.

Soft document frequency. A collection frequency $n_j = \sum_d x_{dj}$ is not a document frequency and breaks Eq. 1: the numerator $N - n_j + 0.5$ goes negative once $n_j > N + 0.5$, routine for a common concept, leaving IDF non-monotone and eventually undefined. We define it through a bounded occurrence gate, $n_j = \sum_d g(x_{dj})$, $g : \mathbb{R}_{\geq 0} \rightarrow [0, 1]$, guaranteeing $n_j \in [0, N]$ and a monotone IDF_j . We compare a hard threshold, a clip $\min(1, u/\tau)$, and a sigmoid, defaulting to the clip.

The S-BM25 score.

$$S(q, d) = \sum_{j \in C_q} w_{qj} \text{IDF}_j \frac{x_{dj}(k_1+1)}{x_{dj} + k_1 \left(1 - b + b \frac{L_d}{\bar{L}}\right)} \quad (2)$$

with $L_d = \sum_j x_{dj}$, $\bar{L}$ the corpus mean, $C_q = \text{supp}(\text{Top-}\kappa_q(x_q))$. The query side, usually left implicit, is made ablatable: $w_{qj} \in \{1, x_{qj}, x_{qj}(k_3+1)/(x_{qj} + k_3)\}$, with query support sparsified independently of the document side, since a dense x_q touches every posting list and dilutes IDF across thousands of weak activations. Top- k sparsification is part of the model, not an efficiency afterthought: it is one of the nonlinearities preventing the collapse below, and Zong et al. (2026) find a token-level top- k constraint improves efficiency and out-of-domain effectiveness even for standard SPLADE.

Proposition 1 (Linear collapse). *Let $\Phi(h) = Ch$ and $S_{\text{lin}}(q, d) = \sum_j \text{IDF}_j x_{qj} x_{dj}$. Then $S_{\text{lin}}(q, d) = e_q^\top C^\top \text{diag}(\text{IDF}) C e_d$, a fixed bilinear form on the original embedding space; if $C^\top \text{diag}(\text{IDF}) C = \gamma I$ it is exactly cosine ranking.*

The proof is immediate and the consequence sharp: $\text{diag}(\text{IDF})$ is absorbed into the matrix, so IDF alone cannot escape the bilinear family and any

Basis	Construction and role
B0 Ident.	Split-rectified raw dims, $K=2d$. Isolates scorer from basis: if S-BM25 here matches B6, concept extraction contributes nothing.
B1 Rand.	Gaussian projection, split-rectified, top- k . Designed <i>low</i> -Zipfianness point: near-uniform frequencies, near-zero IDF variance.
B2 PCA	Top- K' principal directions. Tests whether variance ordering alone yields usable collection statistics.
B3 k -means	K centroids, soft assignment to nearest κ . Expected <i>mid</i> case: centroids track density, equalising occupancy.
B4 PQ	M subspaces $\times 2^b$ codes. Every document activates exactly M concepts, so L_d is constant — a natural experiment on length normalisation.
B5 RQ	L residual levels. The hourglass effect (Zhu et al., 2024) predicts concentration at intermediate levels: high-imbalance-but-not-Zipfian.
B6 SAE	Top- K SAE, $m=32,768$, $k_{\text{SAE}}=16$, FineWeb-Edu. State of the art (Clavié et al., 2026); reproduction target, expected <i>high</i> -Zipfianness.

Table 2: Seven bases, chosen to span a wide range of concept-frequency distributions — which is what makes the H1 regression possible.

real gain must come from a nonlinearity — saturation, input-dependent top- k support, per-document L_d , or a nonlinear Φ or ϕ . This also buys a validity check: on a linear basis the IDF-only, dense-support, no-saturation, no-length configuration is a *predicted null*, and if it beats cosine the tuning protocol is leaking. Efficiency is query–document FLOPs after Zong et al. (2026), with latency under a sparse ANN engine (Bruch et al., 2024).

5 Concept Bases and Diagnostics

Table 2 lists the seven bases. Each also runs at $K \in \{2^{12}, \dots, 2^{17}\}$ where meaningful, turning a 7-point correlation study into one with 30–40 points and sweeping the frequency distribution *within* each family, not only across families.

BM25’s power flows through IDF_j , which can only discriminate if it varies. Natural language supplies that variance free, being approximately Zipfian (Zipf, 1932); a learned vocabulary has no such guarantee. For every (basis, K , granularity), before any retrieval run, we compute: **D1** live-feature fraction $\rho = \frac{1}{K} |\{j : n_j \geq 5\}|$; **D2** $\text{Var}_j[\text{IDF}_j]$; **D3** Zipf slope $\hat{s}$ (OLS of $\log n(r)$ on $\log r$, with R^2); **D4** effective vocabulary K_{eff}/K from Shannon entropy; **D5** Gini of $\{n_j\}$; **D6** head mass $\mu_1\%$; **D7** query–document support overlap, relevant vs. random.

Two design points follow. IDF variance is triv-

ially inflatable — dead latents or empty clusters give enormous $\text{Var}[\text{IDF}]$ and useless retrieval — so D1 gates the rest and configurations are reported as the pair $(\rho, \text{Var}[\text{IDF}])$, never variance alone. And the diagnostics are *not* a GO/NO-GO filter: that BM25-style scoring fails on near-uniform vocabularies can only be established by running retrieval on near-uniform vocabularies.

6 Experimental Design

The factorial. Seven bases $\times \{\text{P}, \text{T}\} \times \{\text{inner product}, \text{S-BM25}\} \times 4\text{--}6$ levels of K , holding encoder, corpus and protocol fixed. The identifying comparison is never S-BM25 against lexical BM25; it is always *same representation, different scoring function*, which licenses attributing a difference to the scorer. Since B6-P with inner product approximates CL-SR and B6-T with BM25 approximates Latent Terms, the 2×2 block at B6 deconfounds the two published results with corpus, backbone and evaluation fixed — the highest-value cell in the study, run first after reproduction.

Encoders and data. Contriever (msmarco), the weaker model on which Latent Terms gains most; nomic-embed-text-v1.5; and GTE-ModernColBERT, to test whether late interaction changes the picture. The full factorial runs on Contriever, the others on a reduced set. Development is on MS MARCO dev. LIMIT (Weller et al., 2026) is the diagnostic set, built to expose single-vector expressivity limits: cosine collapses to ≈ 0.03 R@20 there while lexical BM25 exceeds 0.95, so any recovery is strong evidence the gain is structural. BEIR and TREC-DL 19/20 measure generalisation on the 2–3 best configurations per family only; the full BEIR grid would be $7 \times 2 \times 2 \times 5 \times 15$ evaluations.

Tuning and gate. Hyperparameters are tuned on MS MARCO train/dev; BEIR and TREC-DL stay strictly zero-shot, since Clavié et al. (2026) report per-dataset tuning buys only ≈ 0.009 nDCG@10. We sweep $k_1 \in \{0.9, \dots, 16\}$ (default 8, far outside the lexical range), $b \in \{0, \dots, 1.0\}$, $\alpha \in \{0.25, \dots, 1.0\}$, the three gates, $\kappa_d \in \{16, \dots, 256, \infty\}$, $\kappa_q \in \{8, \dots, 64, \infty\}$ and the three w_q forms. Before any novel configuration is analysed, B6-T with BM25 must reproduce Clavié et al. (2026) within ± 0.015 nDCG (BEIR 0.474 Contriever, 0.526 Nomic; LIMIT R@20 0.510). Failure is blocking, with likely causes in order a BM25 variant mismatch (Kamphuis et al., 2020),

a missing ϕ , or k_1 at lexical defaults. Metrics: nDCG@10, R@100, MRR@10, R@1000; QD-FLOPs, index size, latency; D1–D7.

7 Hypotheses and Analysis Plan

Let Δ_c be the nDCG@10 margin of S-BM25 over inner product for configuration c . We pre-register:

H1 (primary). Δ_c rises with $\text{Var}_j[\text{IDF}_j]$ and with $|\hat{s}|$ approaching 1, controlling for ρ . We regress Δ_c on $\text{Var}[\text{IDF}]_c$, $|\hat{s}_c|$, $(K_{\text{eff}}/K)_c$, ρ_c and $\mathcal{K}[T]_c$ over all $\approx 30\text{--}40$ configurations, with basis-family random effects. The headline is a leave-one-family-out cross-validated R^2 : can a held-out family’s diagnostics predict its margin? Sweeping α within a fixed basis adds a within-basis manipulation — a stronger claim than cross-basis correlation.

H2. Component contribution is conditional on representation: IDF scales with $\text{Var}[\text{IDF}]$; length normalisation acts only under T, inert under P and under PQ where L_d is constant by construction; saturation acts wherever magnitudes are heavy-tailed. Tested by a full 2^4 ablation over {IDF, saturation, length, sparsification} at three best and two worst bases. The PQ prediction doubles as a bug detector.

H3. Granularity, not scoring function, explains the CL-SR/Latent Terms gap. Three outcomes are distinguishable and all informative: granularity dominates, scoring dominates (CL-SR has been misattributed), or they interact (BM25 partly rescues pooled representations).

H4 (null). For linear bases the IDF-only, dense-support configuration does not beat tuned cosine; a “positive” here indicates methodological error, not discovery.

H5. $\Delta_c > 0$ on configurations matching a pre-specified diagnostic profile and $\Delta_c \lesssim 0$ elsewhere, the boundary predictable from diagnostics alone. Uniform wins are not expected and their absence is not failure: the deliverable is a predictive rule, not a leaderboard entry.

Statistics: paired two-tailed t -tests at $p < 0.05$ with Holm–Bonferroni correction within each hypothesis family, bootstrap CIs over 10,000 query resamples, five seeds per SAE, clustering and quantisation run, effect sizes alongside p -values.

8 Timeline, Risks, Contributions

Weeks 1–3 build the framework and index, exiting when lexical BM25 matches Pyserini to three decimals; 4–6 reproduce B6-T within ± 0.015 (gate);

7–8 compute D1–D7 with no retrieval; 9–10 run the B6 2×2 and decide whether pooled bases stay in (gate); 11–15 the full factorial and the H1 fit; 16–18 the 2^4 ablation; 19–21 BEIR, TREC-DL, the remaining encoders and the out-of-family check; 22–23 efficiency; 24–26 testing, feature analysis and writing.

Most failure modes stay informative, which is the point of designing around a correlation rather than a win: poor k -means results, which we expect, are a designed mid-Zipfianness datapoint supporting H1, and pooled granularity failing outright resolves G3. Δ varying while the diagnostics fail to predict it would mean Zipfianness is not the operative variable and the field’s stated explanation is wrong — the most interesting negative available. Two risks are not self-justifying: reproduction failure, which is blocking, and scooping, real at four papers in four months, against which the factorial and the predictive rule rather than the method are the contribution.

The deliverables map onto the gaps: the factorial itself (G1), the Zipfianness hypothesis upgraded from $n=2$ to a regression with out-of-family validation (G2), the pooled/token deconfounding (G3), the first use of quantisation and clustering bases as BM25 vocabularies for text (G4), component attribution including the provable PQ null (G5), plus the collapse result, a predictive rule for choosing a scorer for a new representation, and open artefacts. Two caveats: Zipfianness rests on five correlated statistics, so $\text{Var}[\text{IDF}]$ is pre-registered as primary with the correlation matrix reported; and at 30–40 configurations the regression has modest power, making the K - and α -sweeps the main means of raising n . All work is English and text-only.

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